

INTRODUCTION

Evolutionary Biology is the study of history of life forms on earth. What exactly is evolution? To understand the changes in flora and fauna that have occurred over millions of years on earth, we must have an understanding of the context of origin of life, i.e., evolution of earth, of stars and indeed of the universe itself. What follows is the longest of all the construed and conjectured stories. This is the story of origin of life and evolution of life forms or biodiversity on planet earth in the context of evolution of earth and against the background of evolution of universe itself.

ORIGIN OF LIFE (Biopoiesis)

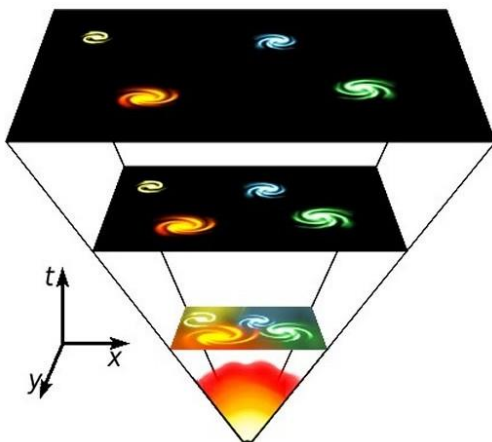
- The origin of life is considered a unique event in the history of universe.
- The universe is very old – almost 20 billion years old.
- Huge clusters of galaxies comprise the universe.
- Galaxies contain stars and clouds of gas and dust.
- Considering the size of universe, earth is indeed a speck.

THE BIG BANG THEORY

- Big Bang theory explains the origin of universe.
- Proposed by **George Lemaitre** in 1927.
- The universe was born in a single titanic explosion called the '**Big Bang**', about 15 billion (15,000 million) years ago.
- There was no galaxies, no stars, and no planets and hence there can be no life.
- All the matter, energy, even space and time, came into being at the precise moment of the big bang.
- The universe expanded and hence, the temperature came down.
- Hydrogen and Helium formed sometime later.
- The gases condensed under gravitation and formed the galaxies of the present day universe



George Lemaitre

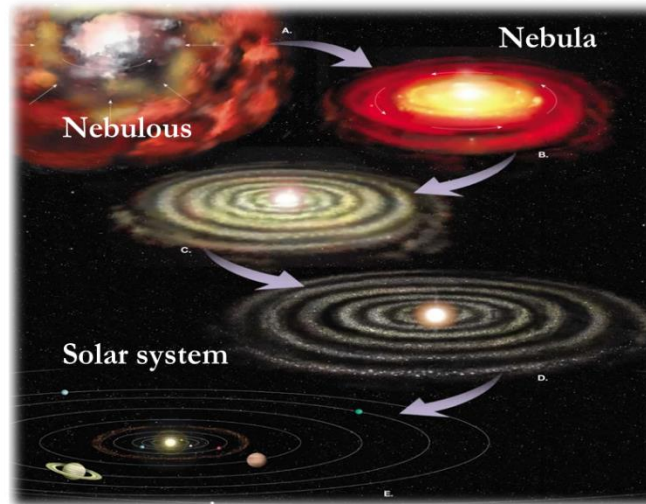


- In the solar system of the Milky Way galaxy, earth was supposed to have been formed about **4.5 billion years back**.
- There was no atmosphere on early earth.
- Water vapour, methane, carbon dioxide and ammonia released from molten mass covered the surface.
- The UV rays from the sun broke up water into Hydrogen and Oxygen and the lighter H₂ escaped.
- The ozone layer was formed
- As it cooled, the water vapour fell as rain, to fill all the depressions and form oceans.
- Life appeared **500 million years** after the formation of earth, i.e., almost **4 billion years** back.



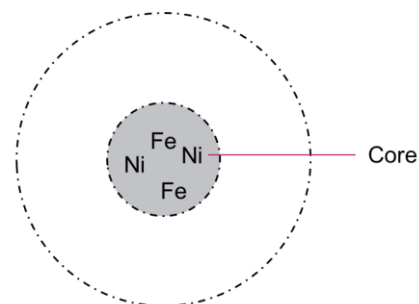
Planet Earth

- Proposed by **Immanuel Kant** in 1755
- According to this theory our solar system originated about 5 billion years back from a rotating cloud of hot gases called nebulous.
- This mass of gas exploded with a big bang into several pieces of hot gases called nebulae.
- One such nebula gave rise to solar system.



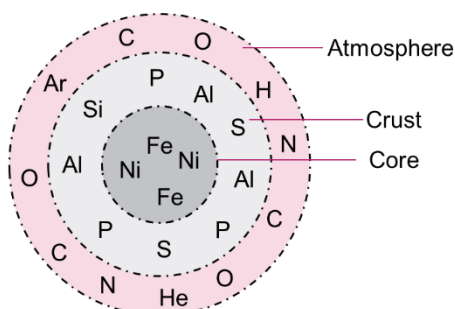
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Random distribution of atoms



Condensation of atoms into distinct layers

- While the **lighter Si, P, S** and **Al** came to the surface and formed **earth's crust**.
- While the **lightest atoms** like **C, H, He, N, O**, and **Ar** escaped the gravitational field and formed the **atmosphere**.



Condensation of atoms into distinct layers



THEORIES OF ORIGIN OF LIFE

1. Theory of Special Creation

- A Spanish Monk, **Father Suarez** (1548 – 1617) proposed the *theory of special creation* based on the Biblical book of **Genesis**.
- According to Genesis, of *Old Testament of Bible*, the world was created by *super natural power* – **GOD**.
- The whole world originated in six natural days.
- Life is considered as a '**vital spirit**' according to this theory.
- **1 & 2nd day** materia prima or earth was created.
- **3rd day** all plants were created
- **4th day** Sun, moon and stars were created
- **5th day** all the aquatic animals and birds were created
- **6th day** animals including human beings were created, first man and then woman.
- First man, **Adam**, and the first woman, **Eve** was created by God



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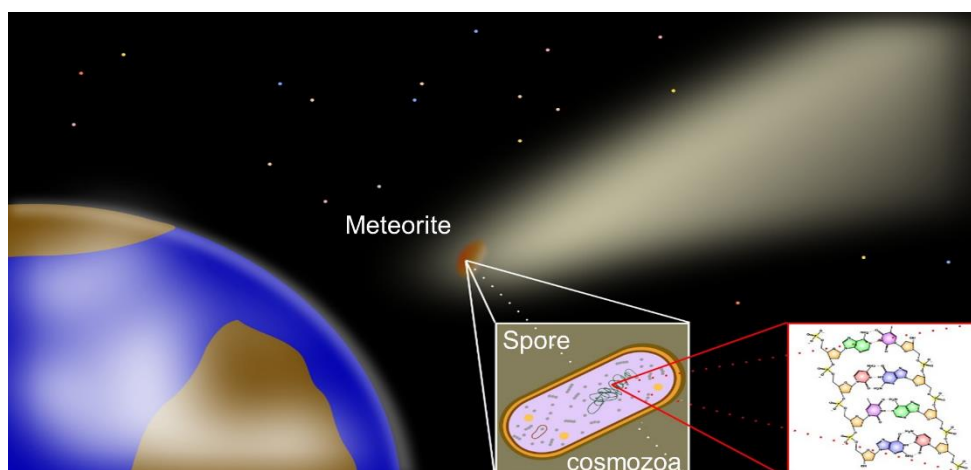


First man, **Adam**,
and the first
woman, **Eve** was
created by God

- This theory is purely based on religious thinking as per the *Genesis of Bible*, and cannot be substantiated by scientific evidences and facts. Consequently, the scientific world has discarded it.

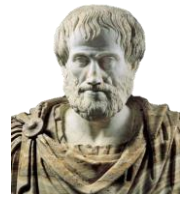
2. COSMOZOIC THEORY (or) PANSPERMIA THEORY

- Developed by **Hermann E. Richter** (1865).
- According to this theory, life on earth has come from distance planet in the form of spore or microorganisms called cosmozoa or panspermia (Gr. pan = all; sperma = seeds) which were preserved alive inside the meteorite.
- When the meteorite struck the earth's surface, cosmozoa were released, which evolved and produced different species that are present today.
- But according to the recent space research programme there is no trace of life on other planets.

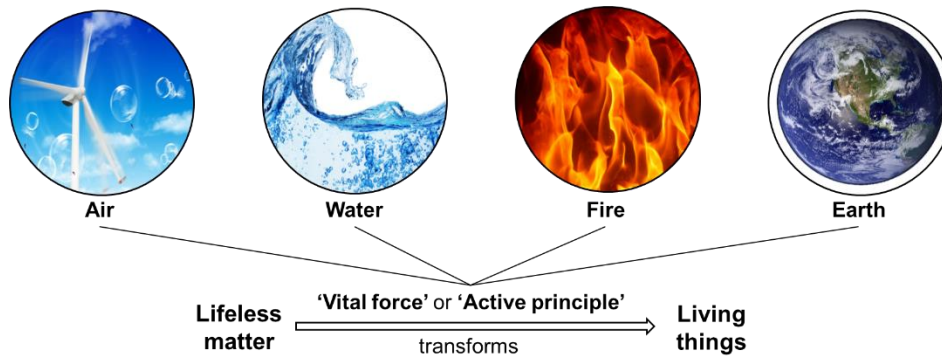


3. THEORY OF ABIOGENESIS (or) THEORY OF SPONTANEOUS GENERATION

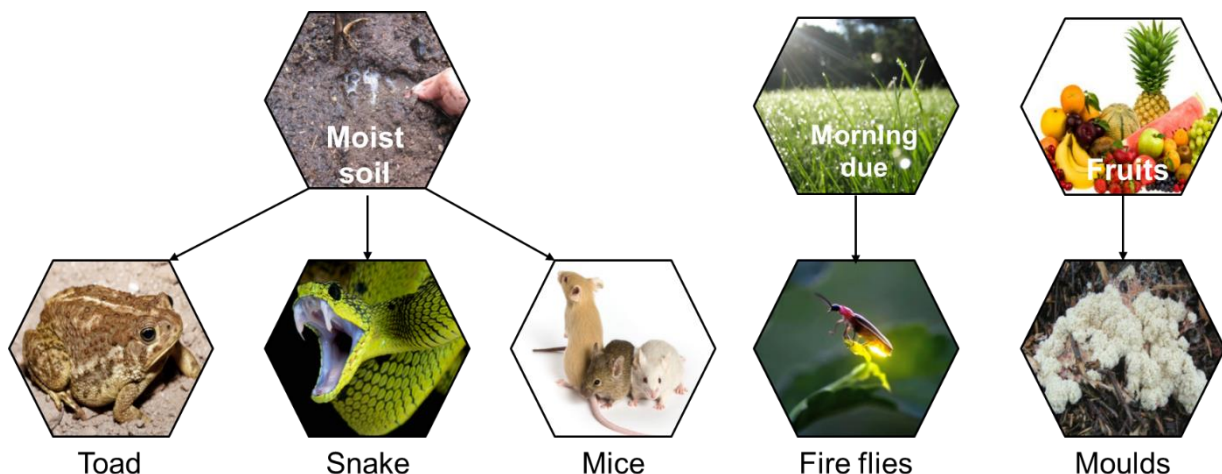
- This was put forth by Greek philosophers in 600 B.C and was supported by Aristotle.
- It is believed that life originated from non-living material spontaneously and continuously.
- They believed that elements such as air, water, fire and earth have 'Vital force' or 'active principle' to transform lifeless matter into living things.



Aristotle



- It was believed that toads, snakes and mice originated from moist soil, fire flies originated from morning dew and moulds originated from fruits, maggots (larvae of house fly) originated from dead flesh and fleas from dust.



Maggots on dead flesh



Van Helmont: (1600s) : He thought he could produce mice from wheat and a soiled shirt.



Dust

Fleas

Recipe for Mice by Dr. Jan Baptista von Helmont

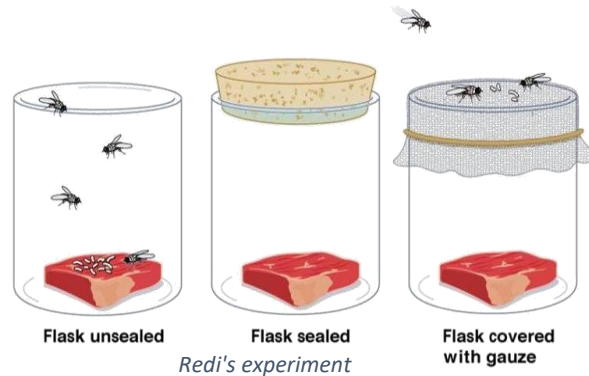


4. THEORY OF BIOGENESIS

- It was put forth by **Francisco Redi**, an Italian biologist and a physician in 1688.
- It was supported by others like **Spallanzani** and **Louis Pasteur**.
- According to this theory, **life originates from pre-existing life by reproduction**.
- For his experiment **Redi** filled a set of jars with meat and covered some jars while he left the others open.
- He observed that maggots did not appear spontaneously.
- But his critics argued that though the maggots did not arise spontaneously the microorganisms did.



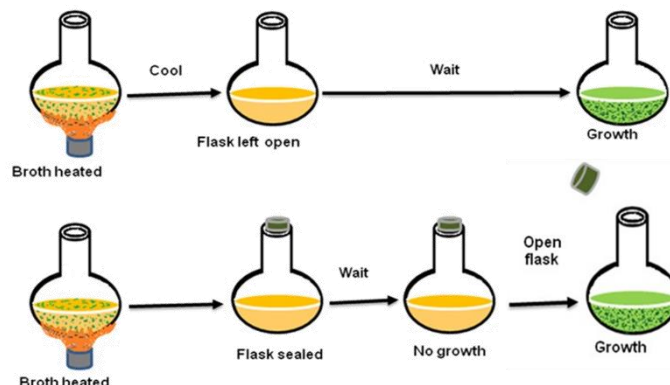
Francisco Redi



- To this criticism in 1765, **Spallanzani** offered a challenging answer through his experiment.
- He boiled the broth (a liquid in which meat, fish, vegetables and grains are cooked) in flask, killed all the microorganism then sealed them
- Critics claimed that boiling had killed the microbes but also drove the vital forces. Hence in the absence of vital forces life could not originate.



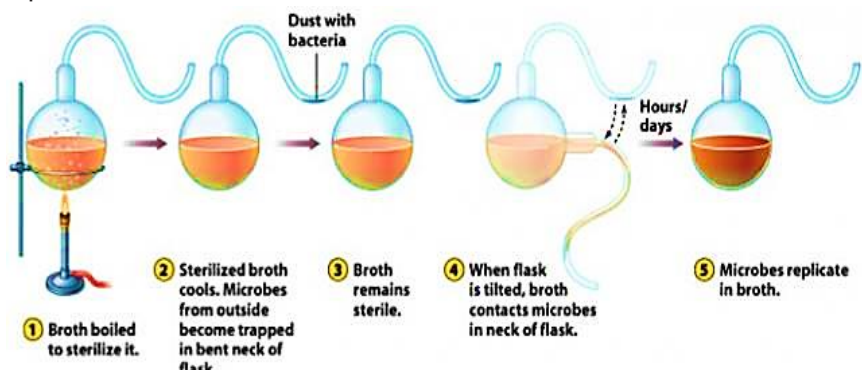
Spallanzani



- **Louis Pasteur** prepared special flasks having long curved neck called '**Swan-necked flasks**' (or S-necked flasks) for the experiment



Louis Pasteur

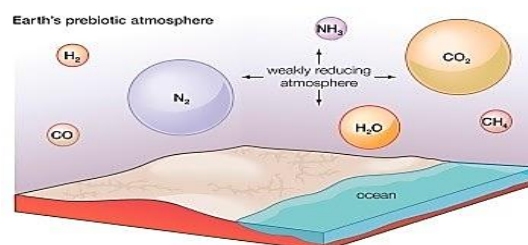
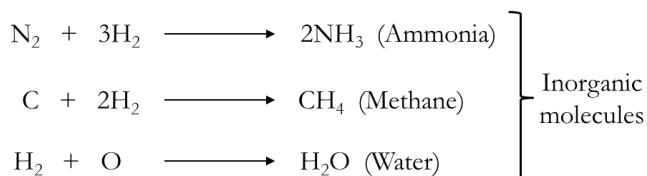


- He took some broth in a round – bottomed flask with a long neck.
- He heated the neck of the flask and gave it an 'S' shaped curvature, which permitted the passage of air to make contact with the broth, while the curved portion trapped the dust and microorganism.

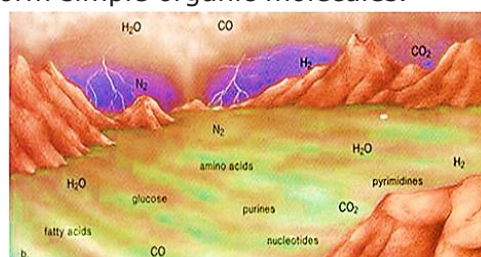
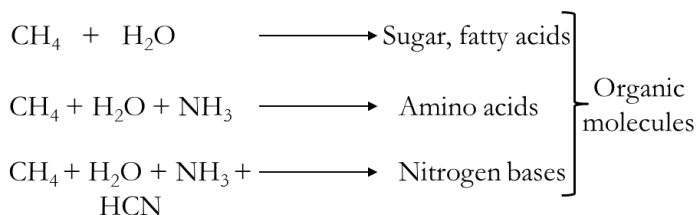
- He boiled the broth to kill the microorganisms and allowed it to cool.
- As the broth remained uncontaminated he proved that even microorganisms arise from pre-existing ones.
- Thus, Pasteur disproved the theory of abiogenesis once for all.
- However, this did not answer how the first life form came on earth.
- To answer the intriguing question as to how the first form of life came into existence scientists pursued different lines of investigation trying to seek an answer to the question of origin of life. The most accepted theory is **Chemical Evolution of life**

5. CHEMICAL EVOLUTION OF LIFE

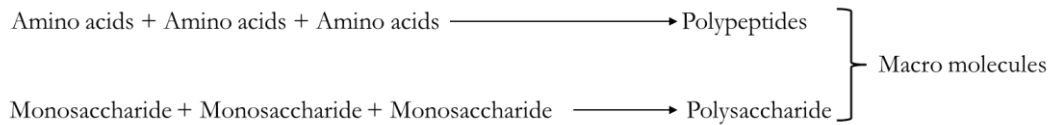
- It is the most widely accepted theory.
- This theory states that, **life originated from chemicals through gradual progressive changes over a long period of time.**
- This theory put forth by two workers independently, a Russian biochemist **Alexander. I. Oparin** (1924) who published it under the title "**The Origin of Life**" and an English biologist, **J.B.S Haldane.**
- This theory suggested that life originated from inorganic substances, which were transformed into organic substances (e.g, RNA, protein, etc.) which in turn were transformed into colloidal systems to produce the first form of life (protobiont / precell)
- Most of the biologist believe that the **prebiotic environment (reducing environment. i.e., there was no free molecular oxygen)** of the earth had very little resemblance to the present day environment.
- Since earth was very hot, the atoms in the atmosphere could not combine with one another to form stable molecule.
- As the earth started cooling gradually, simple atoms in the atmosphere like **H, N, C** and **O** reacted with one another.
- **H** atom being highly reactive combine with the other atoms and formed **simple inorganic molecule** like **water, methane** and **ammonia.**



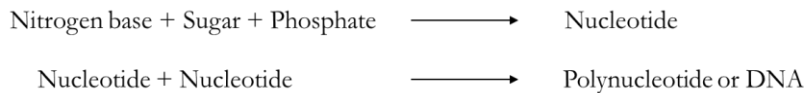
- As the earth's temperature was very high NH_3 and CH_4 remained as gases and water as steam.
- As the process of cooling continued, water vapour in the atmosphere condensed into clouds and it began to rain.
- But, since the earth's surface was still very hot, it could not contain liquid water hence vaporized and returned to the atmosphere. This process continued for millions of years.
- Rain water dissolved atmospheric NH_3 & CH_4 .
- River water carried salts & minerals.
- Volcanic eruption increased mineral content.
- **First step in the origin of life:-** Transformation of atmospheric gases into simple organic molecules
- The availability of energy from Lightening, ultraviolet radiation, volcanic eruption and geothermal heat allowed simple inorganic molecules to form simple organic molecules.



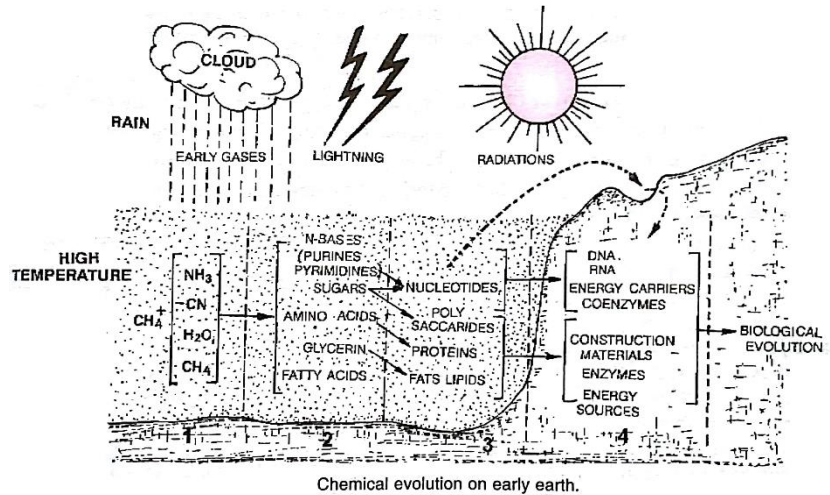
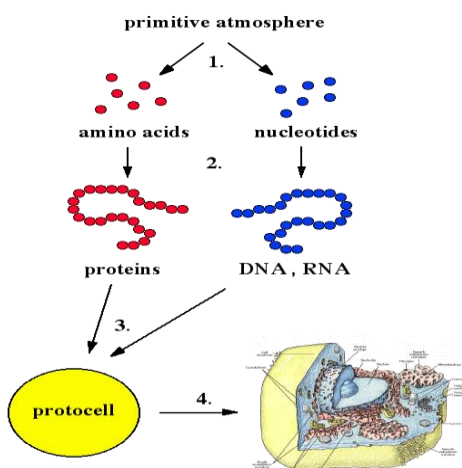
- These micromolecules were in the form of **colloidal droplets**, which did not mix with the surrounding water called '**primitive soup**'.
- Micromolecules began to combine with each another to form larger **macromolecules** by dehydration from the successive building blocks.



- **Next step in chemical evolution was:** - The formation of nucleic acid.



- Thus the first gene was formed which acquired the **self – replicating ability**, a fundamental property of living cell
- The nucleic acids along with other inorganic and organic substance gave rise to the **first form of life** called **protobionts / protocell / ebionts**

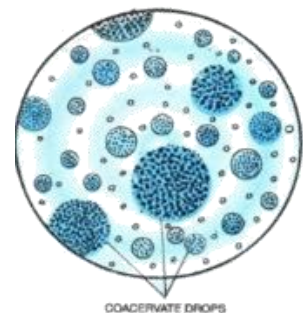


EXTRA INFORMATION

The formation of precell or protobiont is explained through various theories. Some of the important theories are

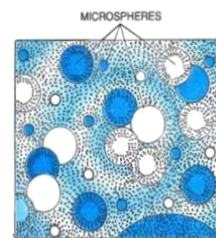
1. Coacervate theory

- It was put forth by Oparin in the year 1938.
- This theory states that protein like substance in the prebiotic period aggregated and developed an enveloping membrane of water.
- These organized structures were called **coacervates** and their formation as **coacervation**.
- Coacervate were non-living, they did exhibit certain structural and functional properties of living organisms and this view is supported by experimental evidences.



2. Microsphere theory of Sydney Fox

- In 1959 **Dr Sydney Fox** elaborated on Oparin's coacervate theory based on his experiment.
- In his experiment he heated the amino acids, which polymerized into protein-like droplets called Proteinoids.
- These thermally produced proteinoids aggregated into spherical colloidal droplets called Microspheres.



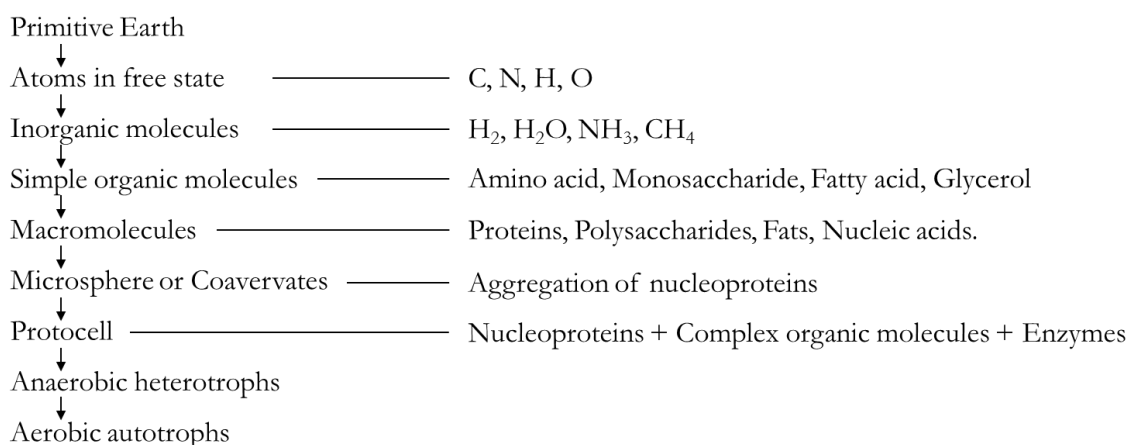
Characteristic features of Microsphere

- Microspheres were bound by a semi-permeable barrier like the living cells.
- They exhibited osmotic property i.e., influenced the passage of materials in and out of the vesicle.
- They also reacted with chemicals like the living cells.
- Had the capacity to replicate by fission and fragmentation.

Thus according to **Oparin** and **Fox**, coacervates and microspheres are the **forerunners of first form of life**.

They were called the **Protocells, protobionts** or **ebionts**. These subsequently gave rise to the **first form of life**. As the pre-cells continued their development, the nucleic acid in the pre-cells not only acquired the ability to replicate, but also directed a series of chemical reactions including **protein synthesis**. This was the significant step in transforming the **pre-cell** into a **cell** and thus gave rise to the first form of life after a long process of **molecular evolution**. With the appearance of the first living organism, **chemical evolution** gave rise to **biological evolution**.

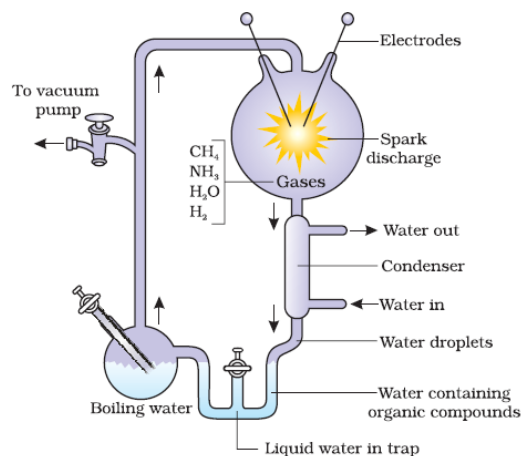
- The first **primitive living organisms** were **heterotrophs**. They began to compete for available food supplies found in the sea and obtained energy by the fermentation of organic substances dissolved in the sea. As they multiplied and flourished, there was a depletion of the existing nutrients as well as an increase in the carbon dioxide concentration due to fermentation. It is during this time **random changes** or **mutation** occurred to enhance the chances of survival. Among the favourable changes was the formation of **chlorophyll molecule**.
- Chlorophyll molecules were then used for photosynthesis and the heterotrophs were transformed into **autotrophs**. Due to the release of oxygen by autotrophs, **reducing environment** was gradually converted into **oxidizing one**. The increase in the atmospheric oxygen and formation of ozone later reduced the ultraviolet rays reaching the earth. This provided stability for the progressive forms of life.



Schematic representation of the significant events during the origin of life

STANLEY MILLER'S EXPERIMENT

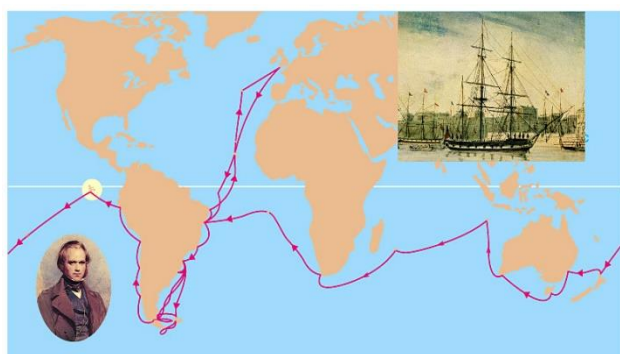
- Oparin-Haldane theory was supported experimentally by **Miller – Urey experiment in 1953** with the help of **simulation experiment**
- Miller and Urey tried to create condition existing in primitive atmosphere – in **Spark Discharge Apparatus** – a large flask containing mixture of **CH₄, NH₃, H₂ and H₂O** with electric source and boiling of water for a week.
- Miller introduced into the apparatus a mixture of gases like **methane, ammonia** and **hydrogen** in the ratio **2 : 1 : 2** (creating a prebiotic environment)
- He partly filled the flask with water and heated it (simulating the ocean).
- This mixture was then exposed to continuous electric discharge (simulating lightning) for gases to interact at 800°C.
- The mixture was then passed through a condenser and collected in the trap.
- Observation:** The experiment yielded several **amino acids** and few other simple **organic compounds** like **HCN, fatty acid, urea** etc.
- Inference:** Thus, Miller's experiment provided vital proof that organic compounds which form the building material of life can be obtained from simple chemicals, prevailed in the prebiotic conditions. Thus, proving the possibility of chemical evolution of life.



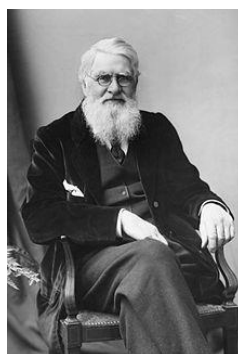
Diagrammatic representation of Miller's experiment

THEORIES OF EVOLUTION

- The theory of special creation or divine intervention was challenged by **Charles Darwin**.
- He made observations on his sea-trip around the world aboard **H.M.S. Beagle** and concluded that all existing living forms share similarities among themselves and also with other life forms, which existed millions of years ago of which many are extinct.
- The evolution of life forms has been gradual and those life forms better fit in environments that leave more progeny. This is called **natural selection** and is a **mechanism of evolution**.
- Darwin proposed natural selection as the mechanism of evolution.
- Alfred Wallace** working in the **Malay Archipelago** also came to the same conclusion.



Charles Darwin sea voyage around world on H.M.S Beagle



Alfred Wallace



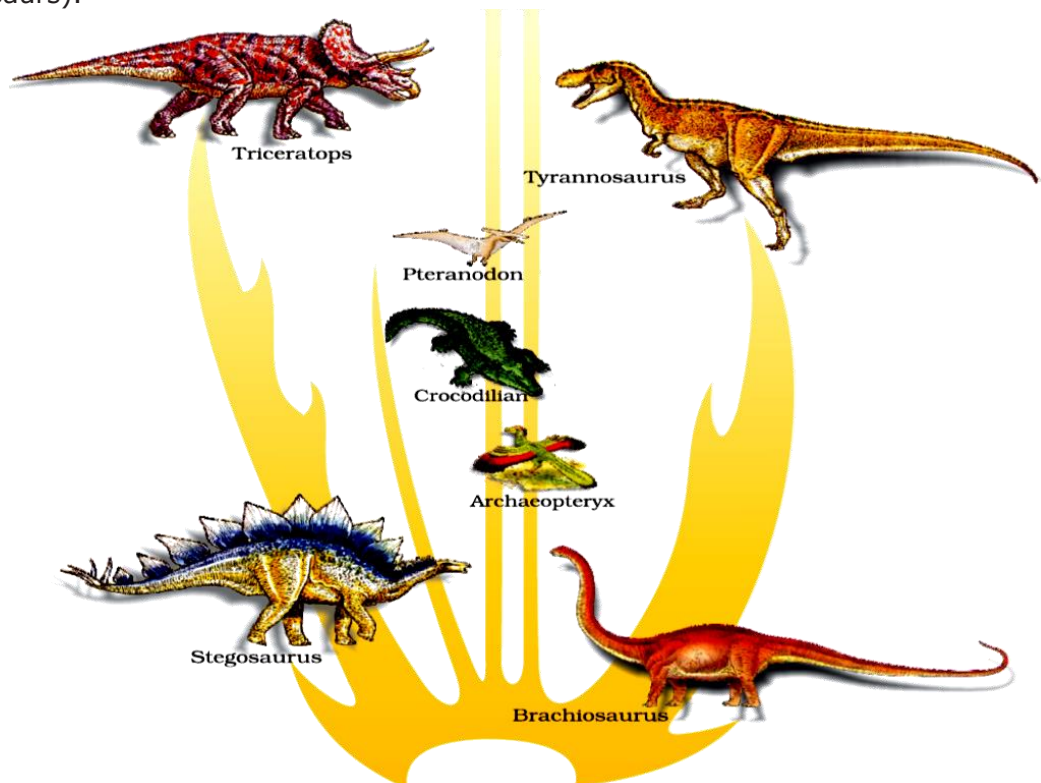
Map showing Malay Archipelago

EVIDENCES OF EVOLUTION

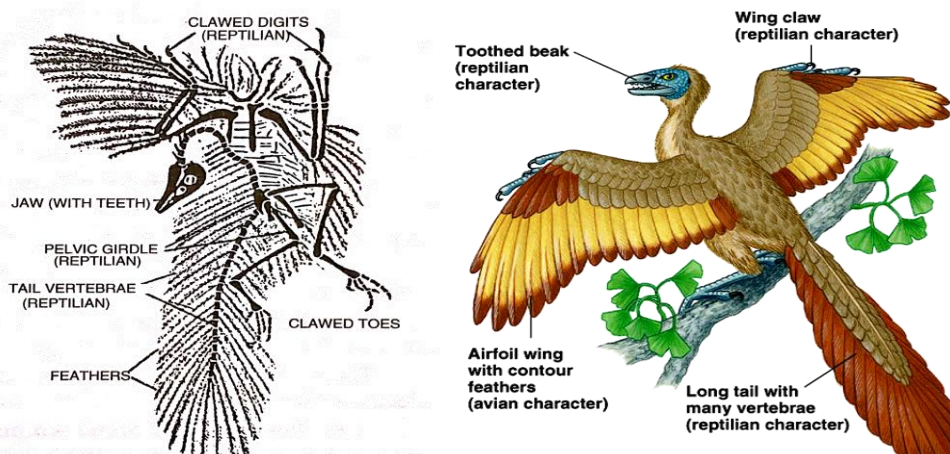
- Paleontological evidences
- Embryological evidences
- Morphological and anatomical evidences
- Biochemical evidences
- Evidences for evolution by natural selection – natural selection by anthropogenic action
- Biogeographical evidences.

1. Paleontological Evidences

- **Fossils** - they represent plants and animals that lived millions of years ago and are now extinct.
- Different aged rock sediments contain fossils of different life-forms, which probably died during the formation of the particular sediment.
- A study of fossils (**Paleontology**) in different sedimentary layers indicates the geological period in which they existed.
- The study showed that life-forms varied over time and certain life forms are restricted to certain geological time-spans.
- Hence, new forms of life have arisen at different times in the history of earth. All this is called paleontological evidence.
- Some of them appear similar to modern organisms. They represent extinct organisms (e.g., Dinosaurs).

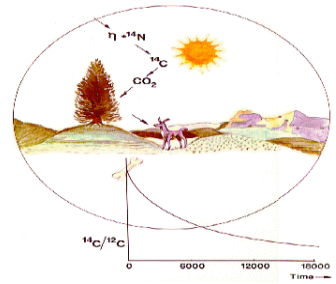


- *Tyrannosaurus Rex* was the largest land reptile which is extinct
- *Pteranodon* was the flying reptile.
- *Archaeopteryx* is the connecting link between reptiles and birds. Reptilian characters in *Archaeopteryx* – teeth in the jaws, claws in the free fingers, unfused tail vertebrae. Avian character in *Archaeopteryx* – jaws modified into beak, and presence of feathers.



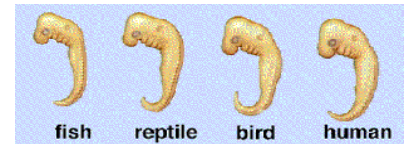
How the Ages of Fossils are calculated?

- The age of fossils are calculated by **radioactive dating** - The technique of comparing the abundance ratio of a radioactive isotope to a reference isotope to determine the age of a material is called radioactive dating.



2. Embryological Evidences

- Biogenetic law** - Proposed by **Ernst Heckel** based upon the observation of certain features during **embryonic stage common to all vertebrates that are absent in adult (Ontogeny recapitulates phylogeny)**
- For example, the embryos of all vertebrates including human develop a **row of vestigial gill slit** just behind the head but it is a functional organ in fish and not found in any other adult vertebrates.
- This proposal was disproved by **Karl Ernst Von Baer**.
- He noted that embryos never pass through the adult stage of other animal



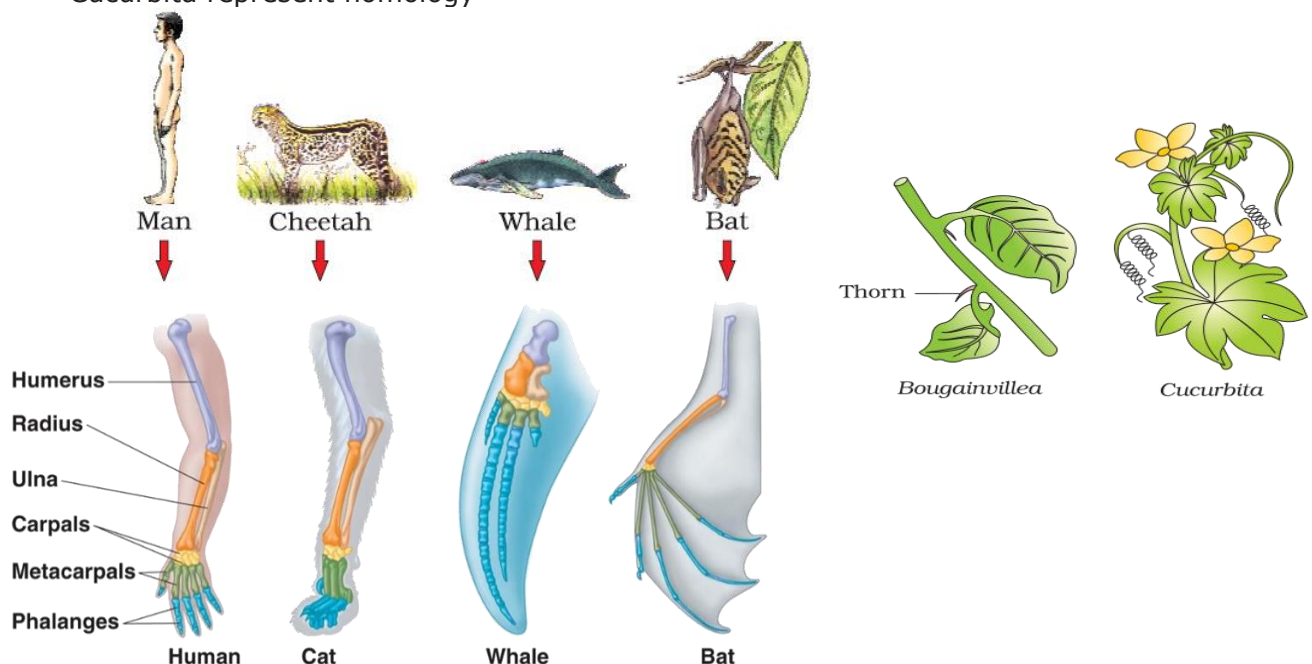
Comparative embryology

3. Morphological and Anatomical Evidences

- It shows evidences of the similarities and differences between living forms of today and that of the prehistoric times. Some of the examples of comparative anatomy and morphology are: Homologous organs and Analogous organs.

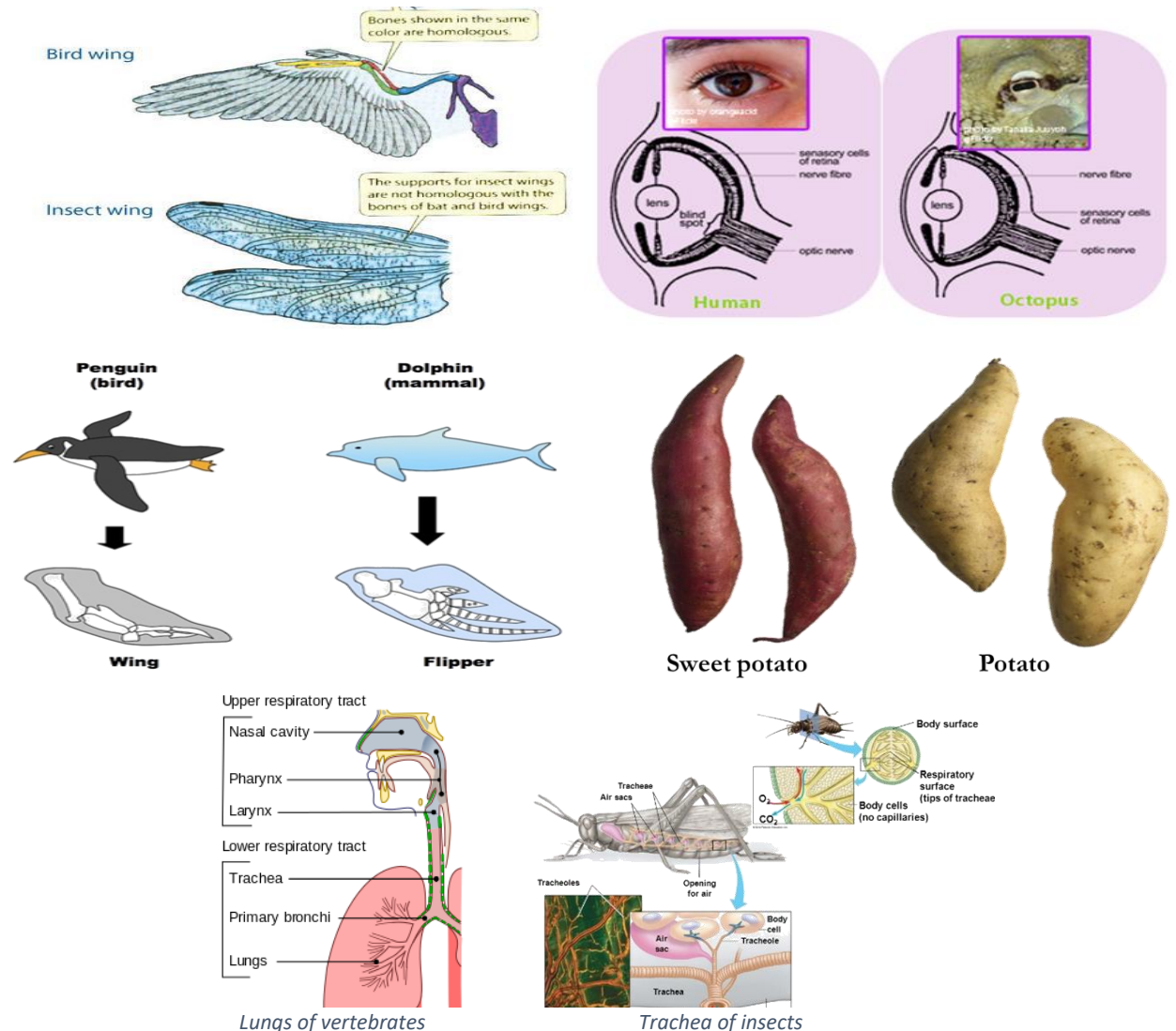
Homologous Organs

- Organs having same anatomical structure but different functions are called **Homologous organs** (common ancestors).
- Homologous organs lead to **divergent evolution** (when two different species evolve differently from a common ancestor but have different characteristics from one another is called divergent evolution)
- Examples –
 - In animals, all mammals share the same pattern of forelimbs. Though they perform different functions, they are anatomically similar.
 - In plants also, the thorn (modified stem) of Bougainvillea and tendrils (modified stem) of Cucurbita represent homology



Analogous organs

- Organs with same function but having different anatomical structures are called as **analogous organs**.
- Analogous leads to **convergent evolution** (when two different species do not share a common ancestor but have developed similar characteristics through adaptation to similar environmental conditions is called convergent evolution).
- Examples –
 - The wings of butterflies and birds
 - Eyes of Octopus (retina from skin) and mammals (retina from embryonic brain)
 - Flipper of penguins and Dolphins.
 - Sweet potato (root modification) and Potato (stem modification)
 - Trachea of insects (from ectoderm) and lungs of vertebrates (from endoderm).
 - Jaws of rabbit (bony) and jaws of cockroach (Chitinous)



4. Biochemical Evidence

- Similarities in proteins and genes.
- Similarities in other biomolecules and metabolism.

5. Evidences for evolution by natural selection - Anthropogenic evolution

- Natural selection is the process by which the organisms that are best suited for their environment survive and reproduce.
- Example – **Adaptive melanism**

Adaptive melanism

- Studied by Fisher and Fort
- In England, it was noted that before industrial revolution (1850s), the number of white-winged moths (*Biston betularia*) was more on tree than that of dark melanised moth (*Biston carbonaria*).
- After industrialization (1920), there were more of dark melanised moths and less white winged moth.



White winged moth and dark winged moth in unpolluted area



White winged moth and dark winged moth in polluted area

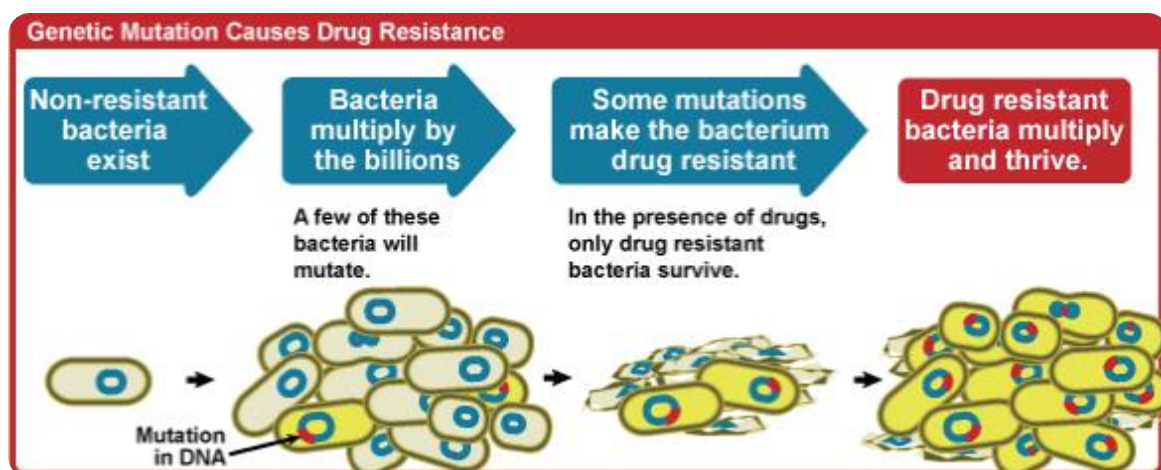
Reasons

- **Before industrialization**
 - There was thick growth of white coloured **lichen** covered the trees.
 - In that background the white winged moths survived but the dark coloured moths were picked out by predators.
- **After industrialization**
 - The tree trunks became dark due to industrial smoke and soots. No growth of lichens.
 - Under this condition the white winged moth did not survive because the predators identified them easily. Dark wing moth survived.



NOTE - lichens can be used as industrial pollution indicators.

Similarly, the herbicide and pesticide resistant plants and animals and antibiotic resistant bacteria are some of the evidences that point towards evolution.



6. Biogeographical Evidences

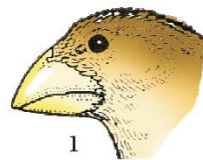
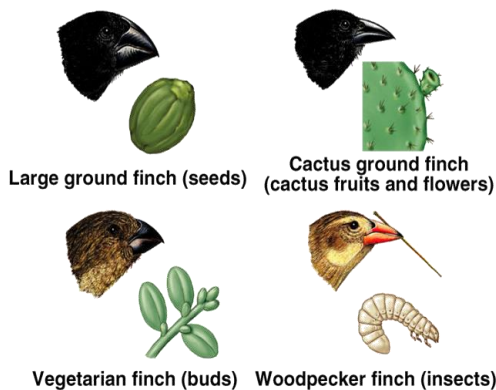
- **Biogeography** is the study of the distribution of species and ecosystems in geographic space and through (geological) time.
- Example – Adaptive radiation of Darwin's finches (seen in Galapagos island), Australian marsupials, Placental mammals in Australia

Adaptive radiation of Darwin's Finches

- During his exploration of the Galapagos Islands, Darwin noticed that there were many varieties of finches in the same island
- They varied from normal seed eating varieties to those that ate insects.
- This process of evolution starting from a single point and radiating in different directions is called **adaptive radiation**.



Charles Robert Darwin
(1809 – 1882)



Cactus-Feeding Finch



Warbler
Finch



Vegetarian Finch

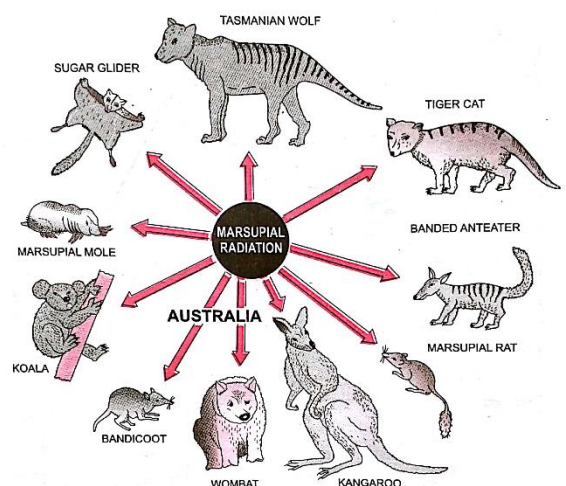
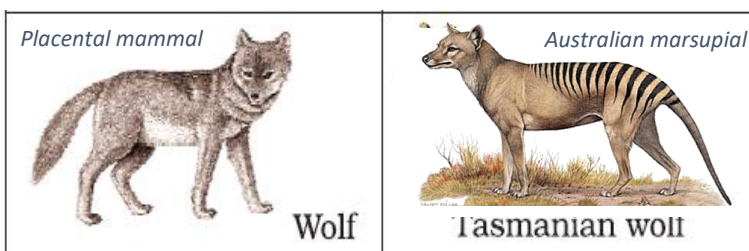


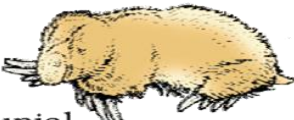
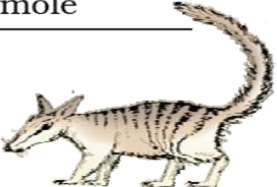
Insectivorous Finch

Variety of beaks of finches that Darwin found in Galapagos Islands

Adaptive radiation of Australian Marsupials

- A number of marsupials, each different from the other evolved from an ancestral stock, but all within the Australian island continent.
 - Placental mammals also exhibit similarities to their corresponding marsupial.
 - Example: placental wolf and the Tasmanian wolf
 - When more than one adaptive radiation occurs in an isolated geographical area (habitat), the phenomenon is called **Convergent evolution**
- 
- The diagram shows three animals: a Sugar Glider on the left, a Tasmanian Wolf in the center, and a Tiger Cat on the right. Red arrows point from each animal towards a common point in the center, illustrating the concept of convergent evolution where different species evolve similar traits independently.



Placental Mammals	Australian Marsupials
 Mole	 Marsupial mole
 Anteater	 Numbat (anteater)
 Mouse	 Marsupial mouse
 Lemur	 Spotted cuscus
 Flying squirrel	 Flying phalanger
 Bobcat	 Tasmanian tiger cat

Picture showing convergent evolution of Australian Marsupials and placental mammals

THEORIES OF BIOLOGICAL EVOLUTION

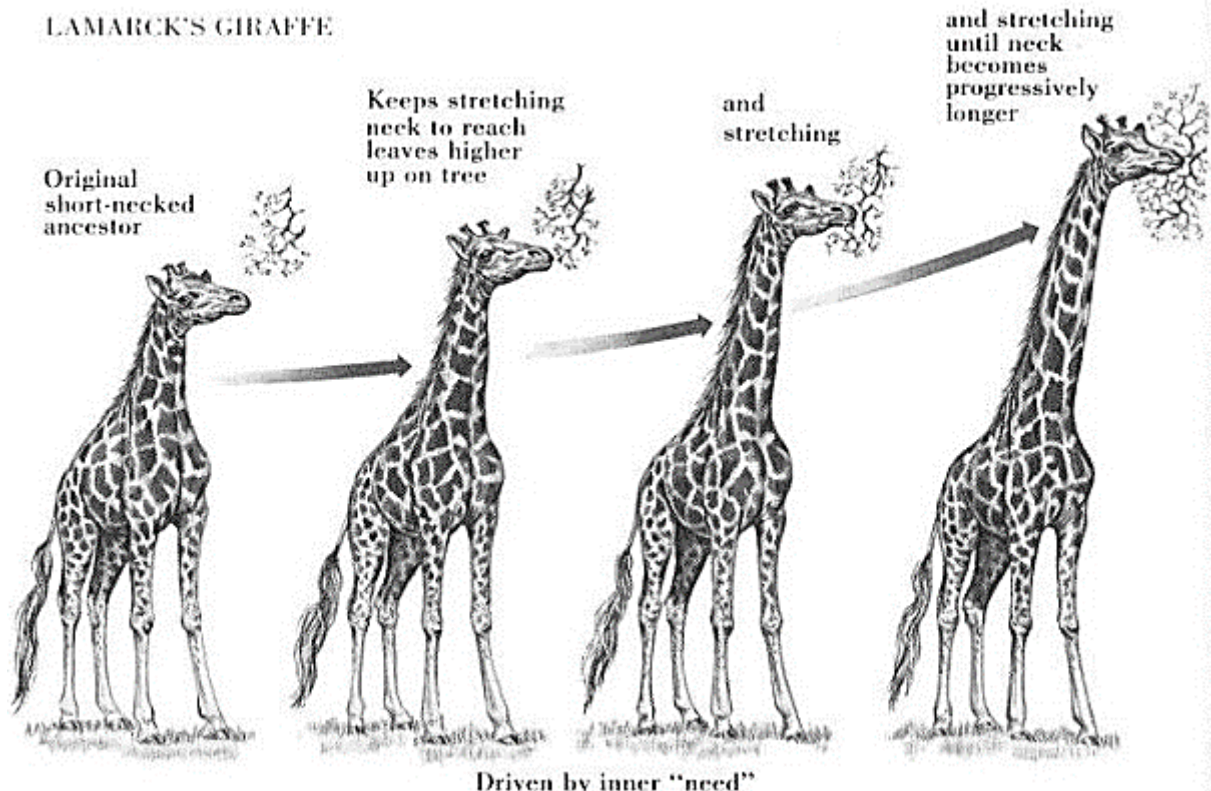
1. Theory of Lamarck (Theory of inheritance of acquired characters / Theory of use and disuse of organs).
2. Darwinian Theory of Evolution / The Theory of Natural Selection.
3. Mutation Theory of Evolution.

NOTE –

- Evolution means opening out or unfolding. It explains the formation of non-living things evolution-of-man and living things.
- The term organic evolution means continuous change or descent with modification or the developmental stages of an organism repeats the developmental stages of ancestors.
- The organic evolution can be defined as the formation of complex organism from simple organism over a period of time. The raw materials for evolution are variations.

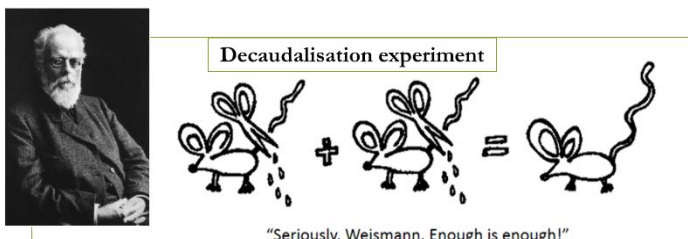
1. Theory of Lamarck (Theory of Inheritance of Acquired Characters / Theory of use and disuse of organs)

- This theory was proposed by French biologist, **Jean Baptist Lamarck**.
- He was the **first scientist to explain the organic evolution**.
- Lamarck published his principle in the book, **Philosophic Zoologique** in 1809.
- **The principles of Lamarck are** –
 - Influence of the environment
 - Use and disuse of organs
 - Inheritance of acquired characters
- It states that evolution of life occurred by use and disuse of organs.
- **Evolution by use of organs:** giraffe have developed long necks as a result of attempts to eat leaves high up on trees.
- **Evolution by disuse:** total disappearance of limbs in snakes.



- Such acquired characters inherit to succeeding generations
- This theory was eliminated out because it is proved that the characters are inherited only through genes.

Decaudalisation experiment:



- It was conducted by **August Weismann** to disprove the Lamarck's inheritance of acquired characters. He collected two rats and cut their tails. Weismann crossed such tail less (acquired character) rats and observed the progeny. He got the tailed progeny. He repeated the experiment (cutting of tails, crossing and observation

of progeny) upto 22 generations. Weismann did not get not even a single rat without tail. This showed him that the acquired character tail lessness is not transmitted to the progeny. So, the Weismann is considered as a critic of Lamarck.

- The above experiment resulted the Weismann to propose Germplasm theory.

Germplasm theory:

- It was proposed by Weismann. The body of every individual contains two plasms. They are somatoplasm and germplasm. Somatoplasm is found in all organs except gonads. Germ plasm is found only in gonads. Germplasm theory states that the changes of germplasm are only transmitted to the progeny. But somatoplasmic changes like tailless, holes in the nose are never transmitted to the progeny. (**Inheritance only takes place by means of the germ cells** – the gametes such as **egg cells** and **sperm cells**. Other cells of the body – **somatic cells** do not function as agents of heredity)

Germplasm theory-experiment:

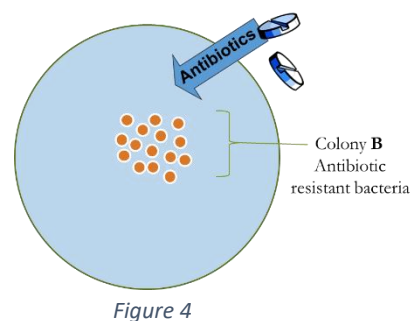
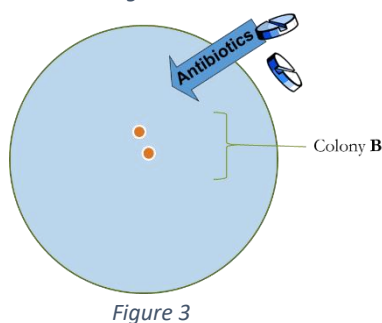
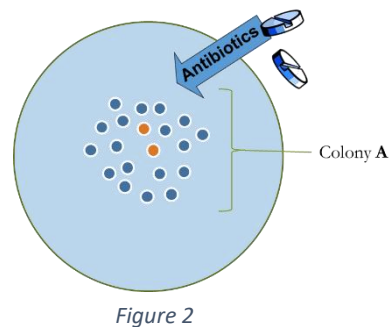
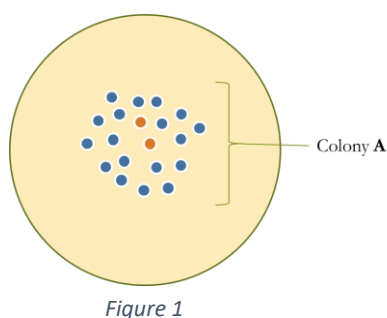
- It was conducted by **Castle** and **Philips** on Guinea pig. They transplanted the ovaries of homozygous black Guinea pig to the white female. Then Castle and Philips crossed such female white Guinea pig with white male. They got heterozygous black progeny. This indicates that the germplasm change of white female is transmitted to the progeny.
- Even though the Lamarckism was highly criticised, still some scientists believe in Lamarckism. They are called neo-Lamarckists. They are Paul Kammerer, Guyer, Smith and McDugall.

2. Darwinian Theory of Evolution / The theory of Natural Selection

- Proposed by Darwin and it is based on **natural selection** and **branching descent**.

Natural selection

- Consider a **bacterial colony A** growing on a given medium. If the medium composition is changed, only a part of the population (say **B**) can survive under new condition. This variant population outgrows the other and appears as new species. i.e., **B is better than A under new condition**. Nature selects for fitness.



Natural selection is based on the following facts

- Overproduction by organisms
- Limited natural resources
- Struggle for existence
- Variations
- Survival of the fittest (by **Herbert Spencer**) / Natural selection (by Darwin)
- Inheritance of useful variations
- Formation of new species

Extra information

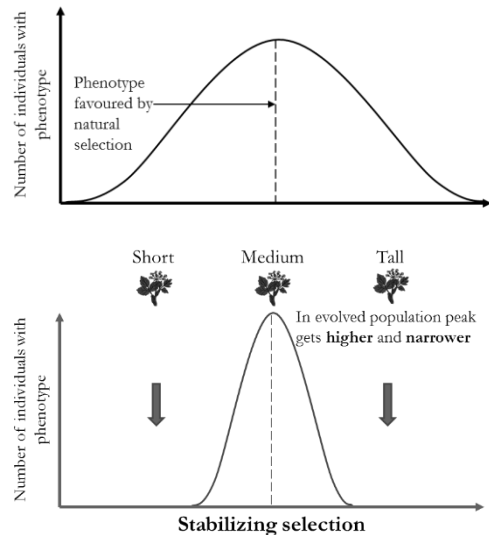
The work of **Thomas Malthus** on population influenced Darwin. Thomas Malthus published the book **An Essay on the Principles of Population** in 1798. He stated that population grows geometrically whereas food grows only arithmetically. Darwin published the book **Origin of Species**.

Types of Natural Selection

1. Stabilizing selection
2. Directional selection
3. Disruptive selection

1. Stabilizing selection – Example - Plant height

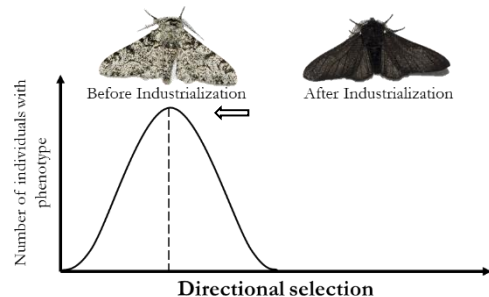
- A plant that is too short may not be able to compete with other plants for sunlight.
- However, extremely tall plants may be more susceptible to wind damage.
- Combined, these two selection pressures select to maintain in plants of medium height.
- The number of plants of medium height will increase while the number of short and tall plants will decrease.



2. Directional selection – Example – Adaptive melanism

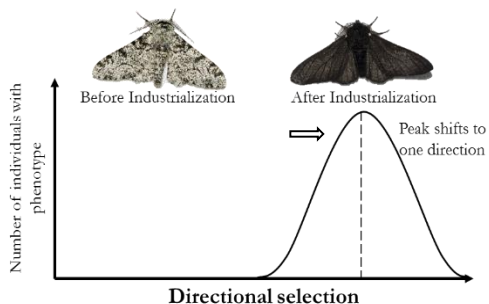
Before industrialization

- There was thick growth of white coloured **lichen** covered the trees.
- In that background the **white winged moths** survived but the **dark coloured moths** were picked out by predators.
- Hence, **increase in the number of the white winged moths**.
- Thus **peak shifts one direction** i.e., **towards white winged moths**.



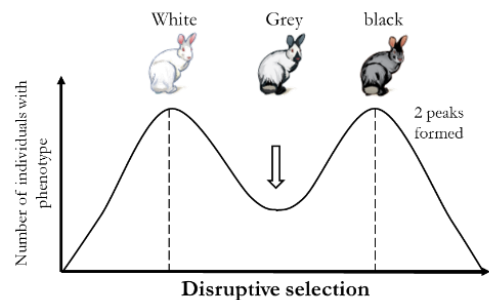
After industrialization

- The tree trunks became dark due to industrial smoke and soots. **No growth of lichens**.
- Under this condition the **white winged moth did not survive** because the predators identified them easily. **Dark winged moth survived**.
- Hence, **increase in the number of the dark winged moths**.
- Thus **peak shifts one direction** i.e., **towards dark winged moths**.



3. Disruptive selection – Example – Fur colour of Rabbit

- In a population of Rabbits there **White coloured, Grey coloured and black coloured rabbit**.
- If this population of rabbits occurred in an environment that **had areas of black rocks as well as areas of white rocks**, the rabbits with black fur would be able to hide from predators amongst the black rocks, and the rabbits with white fur likewise amongst the white rocks.
- **The rabbits with grey fur**, however, would stand out in all areas of the habitat, and would thereby **suffer greater predation**.



Criticism of The Natural Selection Theory

The natural selection theory does not explain the following

- **Inheritance of small variation** which are not useful for the possessor.
- **Over-specialization** of some organs such as tusk of elephant and antlers of deer.
- **Vestigial organs** are present in some animals when they have no function?
- **Arrival of fittest** is not explained by this theory
- **Degeneration of certain organs** in animals is not explained by this theory
- **Discontinuous variations** the theory fails to explain the cause of sudden changes in the body.
- **The main drawback of Darwin's theory was lack of the knowledge of the heredity and that is why he could not explain that how the variations are caused.**

3. Mutation Theory of Evolution

- Proposed by **Hugo de Vries** (1901)
- Based on his work on **evening primrose** (*Oenothera lamarckiana*) suggested that **variations occurred due to mutations**.
- Hugo de Vries gave the name **Saltation (single step large mutation)** to the mutations which brought about speciation.
- **Mutations are random and directionless** while the variations that Darwin talked about were small and directional.

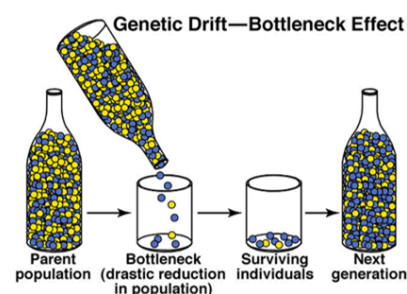
Comparison between Darwinian variation and mutation	
Darwinian variation	Mutation
Minor variation	Large variation
Directional	Random, sudden and directionless
Gradual evolution	Speciation by saltation (single step, large mutation)

Hardy – Weinberg Principle

- The frequency of occurrence of alleles of a gene in a population remains constant through generations unless disturbances such as mutations, non-random mating, etc. are introduced.
- Genetic equilibrium (gene pool remains constant) is a state which provides a baseline to measure genetic change.
- Sum total of all allelic frequencies is 1 ($p + q = 1$).
- Individual frequencies are represented as p and q such as in a diploid, where p and q represent the frequency of allele A and a.
- The frequency of AA is p^2 , that of aa is q^2 , and that of Aa is $2pq$.
- Hence, **$p^2 + 2pq + q^2 = 1$** , which is the expansion of $(p + q)^2$.
- When the frequency measured is different from that expected, it is indicative of evolutionary change.

Hardy – Weinberg equilibrium is affected by

- Gene flow or gene migration
- Genetic drift (changes occurring by chance due to non-random mating)
- Mutation
- Genetic recombination
- Natural selection



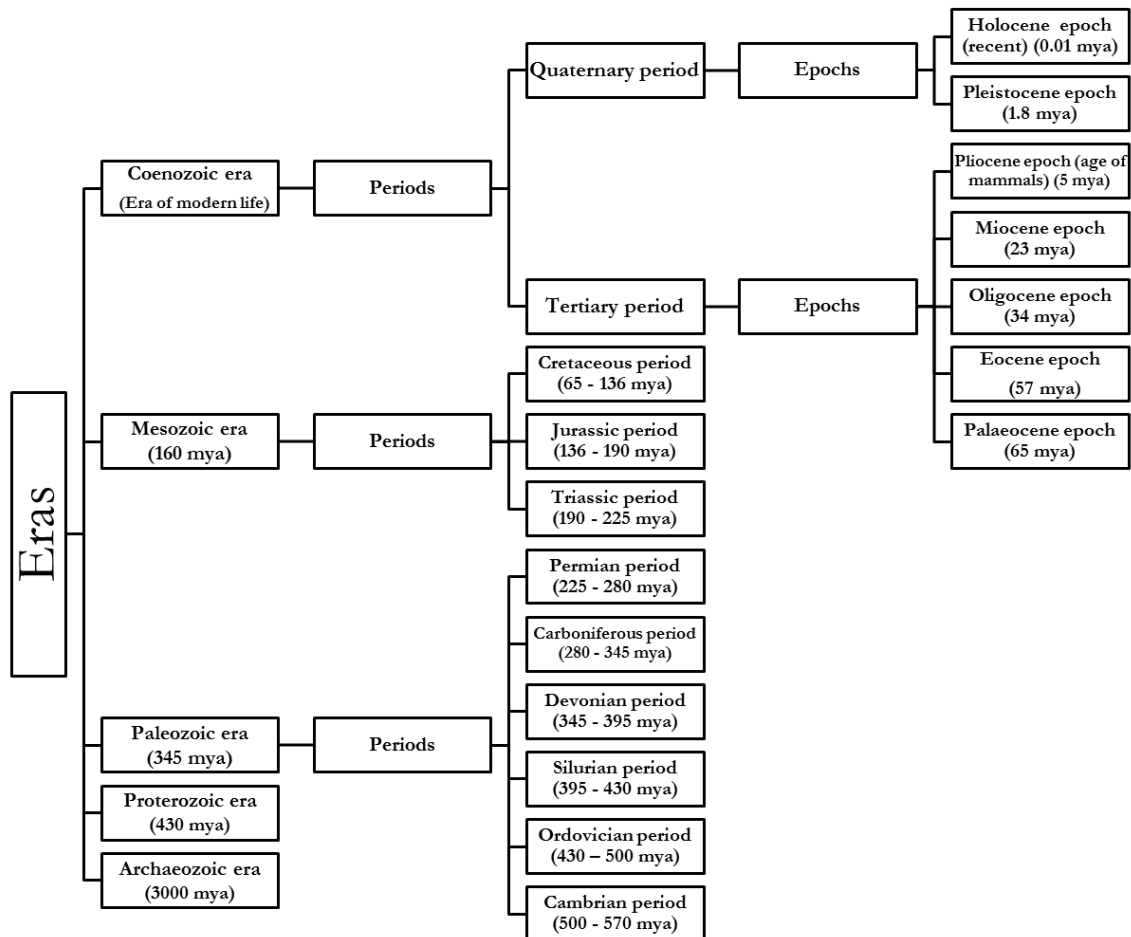
NOTE - Sometimes, the change in allele frequency is so prominent in the new sample of population that they become a different species and the original drifted population becomes the founder. This effect is **called founder effect (Bottle neck effect / Sewall Wright effect)**. The advantageous mutations that help in natural selection over the generations give rise to new phenotypes and result in speciation.

Geological time scale – the whole past history of the earth.

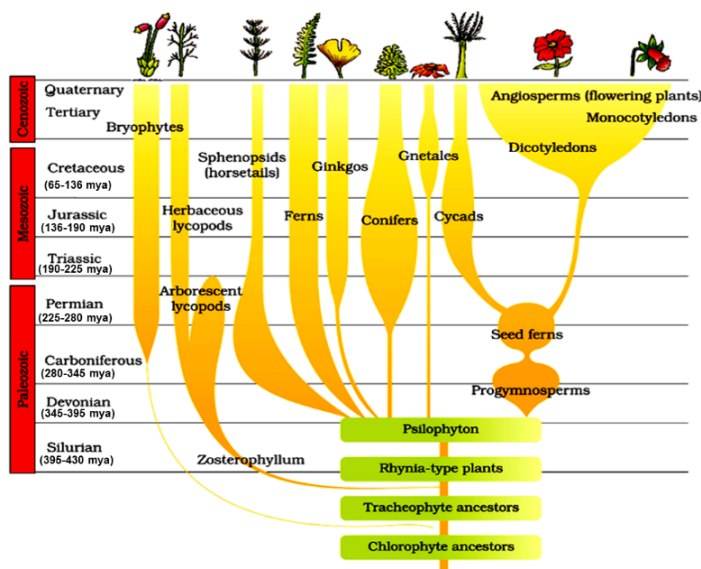
- It was first developed by **Giovanni Arduina** (Geologist) an Italian scientist in 1700.
- The whole past history of the earth is divided into **5 Eras, Periods** and **Epochs**.

Five Division of ERA

1. **Archaeozoic era**
2. **Proterozoic era** } Precambrian era (do not contain any fossils)
3. **Paleozoic era**
4. **Mesozoic era**
5. **Cenozoic era** – most recent and present existing one



EVOLUTION OF PLANTS

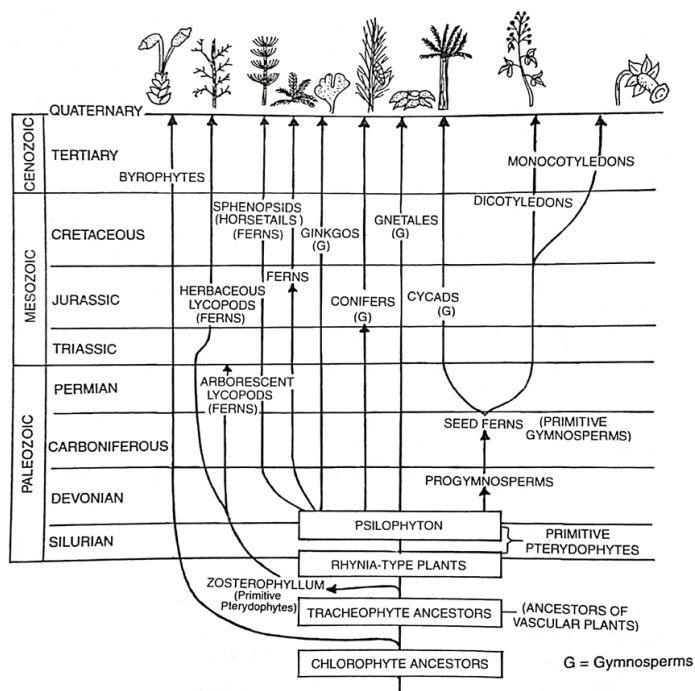


- Cellular life forms occurred on earth about 2000 million years ago (mya).
- Some of these cells had the ability to produce oxygen through reactions similar to photosynthesis.
- Slowly, single-celled organisms became multicellular.
- Seaweeds and some plants probably existed around 320 million years ago.

Extra Information for NEET

Evolution of Plants

- Eukaryotes further diversified to form green algae and early invertebrates.
- The first organisms that existed on land were plants.
- Bryophytes were the first plants to colonise lands.
- Fossils in good number are found from **Palaeozoic era**. Middle Palaeozoic is also called "**age of algae**". Late Palaeozoic is called "**age of ferns**".
- All algal groups became established during **Cambrian period**.
- First land plants** (psilophytes) originated in **Ordovician period**. Marine algae were abundant in Ordovician period
- Earliest spore bearing plants were developed in **Silurian period**. Origin of vascular plants (gymnosperms and angiosperms) took place in this period. It is also **period of origin of ferns**.
- Herbaceous lycopods (ferns) and arborescent lycopods (ferns) evolved from **Zosterophyllum** of palaeozoic era.
- Psilophyton* is the common ancestor for horsetails, ferns and gymnosperms.
- Devonian** - Earliest mosses and ferns
- Carboniferous** - Abundance of tree ferns forming coal forests. First seed plants appeared.
- Mesozoic era** - Also called age of gymnosperms.
- Permian** - Origin of conifers.
- Triassic** - Abundance of ferns, cycads and conifers.
- Jurassic** - Age of cycads (they were in abundance). Origin of angiosperms also took place in this period.
- Ferns and gymnosperms began to decline during **cretaceous period**. There was dominance of flowering plants.
- Tertiary period** is also called **age of angiosperms**. **Quaternary period** is similarly known as the **age of herbs**.
- Eocene** - Angiosperm dominance increase.
- Oligocene** - Rise of monocots.
- Miocene** - Adaptive radiation of angiosperms
- Pliocene** - Adaptive radiation of flowering plants.
- Holocene** - Development and rise of herbs, decline of woody plants.



A sketch of Evolution of Plants through geological periods.

Evolution of Animal

- Animals evolved about 500 million years ago. The first of them to evolve were invertebrates.
- Jawless fishes (petromyzon) evolved around 350 million years ago.
- Some of the fishes (*Ichthyostega* - connecting link between fishes and amphibians) could go on land, and then come back to water. These were the first amphibians.
- In 1938, a fish Coelacanth (ancestor of modern day frogs and salamanders) which was thought to be extinct, was caught in South Africa in 1938. This variety of fish, called lobefins, is believed to have evolved into the first amphibians.
- Amphibians evolved into reptiles.
- In the next 200 million years, reptiles of different sizes dominated the earth.
- The land reptiles were, of course the dinosaurs.
- The biggest of them, i.e., *Tyrannosaurus rex* was about 20 feet height and had huge fearsome dagger like teeth

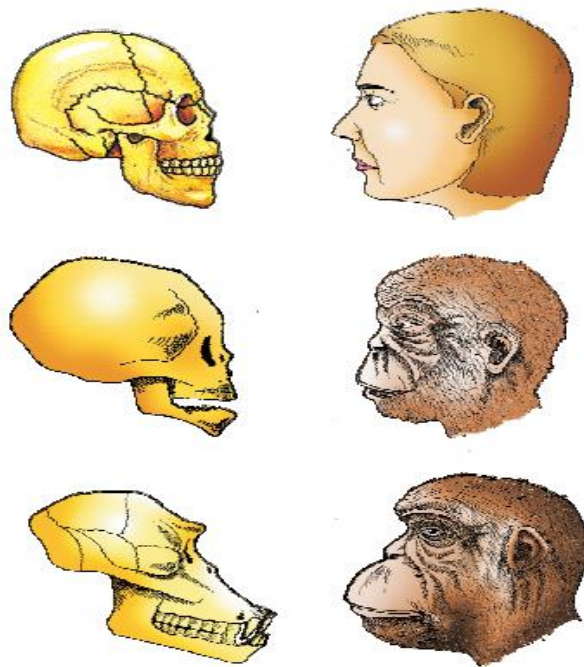
-
- The diagram illustrates the evolutionary relationships between various groups of animals over geological time. The vertical axis represents time in millions of years, from 0 (Quaternary) to 350 (Carboniferous). The horizontal axis represents evolutionary lineages. Key groups shown include Turtles, Lizards, Snakes, Tuataras, Crocodiles, Birds, and Mammals at the top. Extinct groups shown include Dinosaurs, Therapsids, Thecodonts, Pelycosaurs, Early reptiles, and various extinct synapsids. The tree illustrates the divergence of lineages and the extinction of many groups over time.
- Geological Time Scale (Millions of Years):**
- Quaternary: 0
 - Tertiary: 50
 - Cretaceous: 100
 - Jurassic: 150
 - Triassic: 200
 - Permian: 250
 - Carboniferous: 300, 350
- Evolutionary Lineages and Groups:**
- Anapsids:** Includes Turtles.
 - Diapsids:** Includes Lizards, Snakes, Tuataras, Crocodiles, and Birds.
 - Sauropsids (Amniotes):** Includes Lizards, Snakes, Tuataras, Crocodiles, and Birds.
 - Synapsids:** Includes Mammals and various extinct groups like Pelycosaurs, Thecodonts, Therapsids, and Dinosaurs.
 - Primitive synapsids are usually called pelycosaurs.**

- **Therapsida** – cynognathus – was more advance mammals like reptile and was ancestor of present day mammals.
- **Cambrian period** – trilobites (Arthropods) were dominant and became extinct in Permian period
- **Devonian period** – age of fish and amphibians origin,
- **Carboniferous period** – age of amphibians (amphibians dominated)
- **Mesozoic era** – golden age of reptiles
- **Triassic period** – Origin of Dinosaurs, radiation of reptiles, origin of mammals like reptiles
- **Jurassic period** (age of reptiles) – origin of birds,
- **Cretaceous period** – dinosaurs became extinct

- **Pleistocene epoch** – primitive man (*Australopithecus* – first ape man – lived in a cave – was erect posture, omnivore and cranial capacity – 500-700cc) originated
- **Coenozoic era** – age of mammals
- **Over all** – Paleozoic era – age of fishes and reptile,
– Mesozoic era – age of reptiles
– Coenozoic era – age of mammals

ORIGIN AND EVOLUTION OF MAN

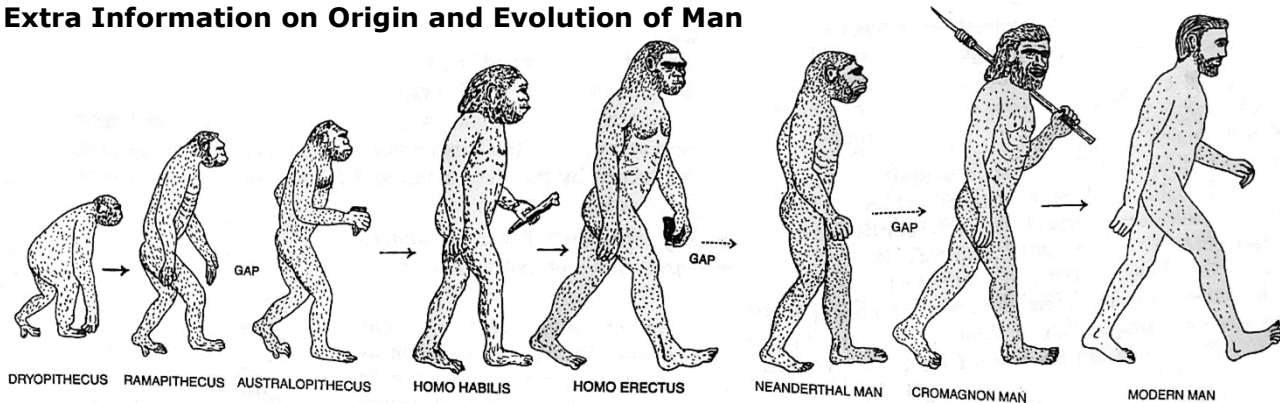
- About 15 mya, primates called *Dryopithecus (Proconsul)* and *Ramapithecus* were existing. They were hairy and walked like gorillas and chimpanzees.
- *Ramapithecus* fossils were found in Shivalik hills and was more like man-like while *Dryopithecus* fossil discovered in Africa and Europe and was more ape-like which is believed to be common ancestor of man and apes.
- Few fossils of man-like bones have been discovered in Ethiopia and Tanzania. These revealed hominid features leading to the belief that about 3-4 mya, man-like primates walked in eastern Africa. They were probably not taller than 4 feet but walked up right.



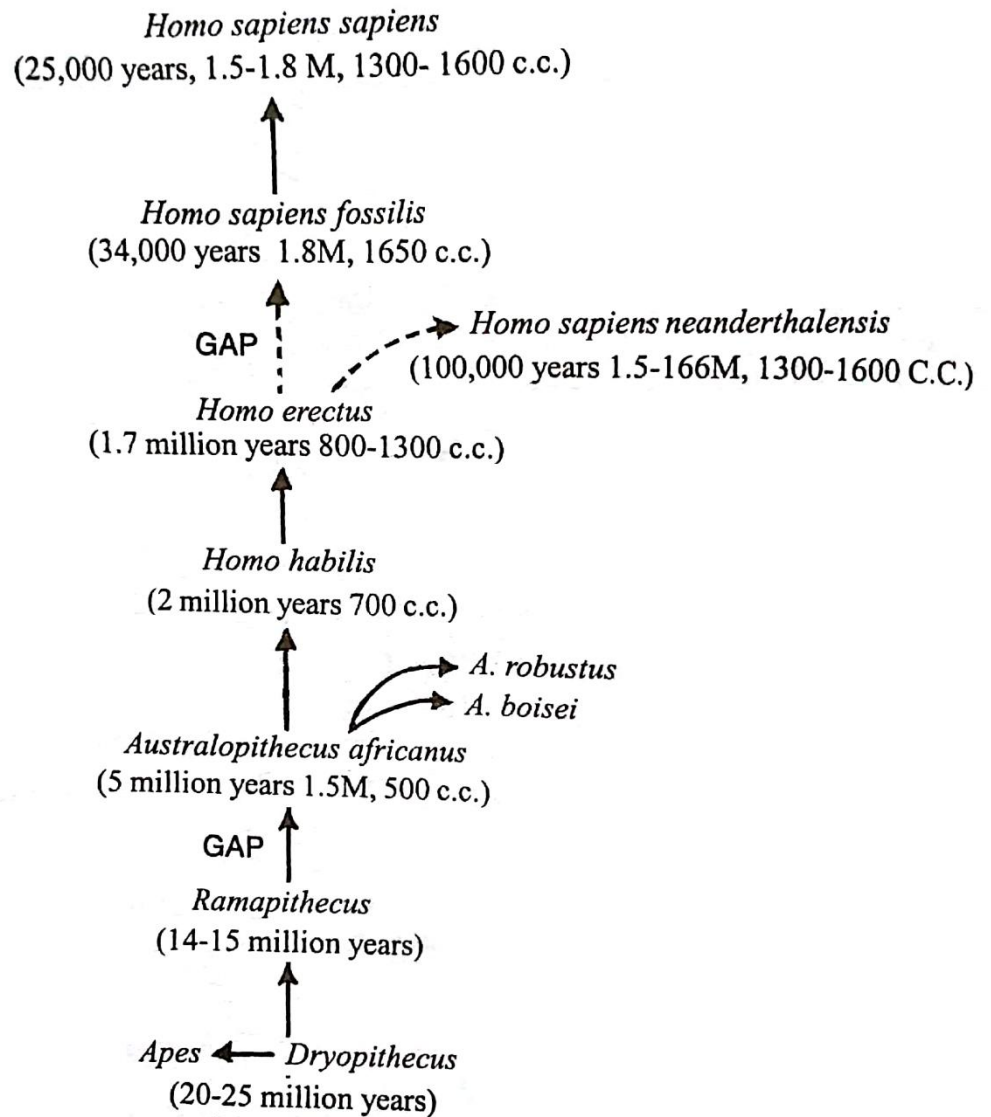
*A comparison of the skulls of adult modern human being, baby chimpanzee and adult chimpanzee.
The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull.*

- 2 mya, *Australopithecines* probably lived in East African grasslands. They hunted with stone weapons but essentially ate fruit. This was the first human like being hominid and was also called as *Homo habilis* (Handy man). The brain capacities were between 650-800cc. They probably did not eat meat.
- Fossils discovered in Java in 1891 revealed the next stage, i.e., *Homo erectus* lived about 1.5 mya and had a large brain capacity around 900cc. They ate meat.
- *Homo erectus* includes three fossils – *Pithecanthropus erectus* (java ape man), *Sinanthropus pekinensis* (Peking man) and Heidelberg man.
- The Neanderthal man with a brain size of 1400cc lived in near east and central Asia between 1,00,000 – 40,000 years back. They used hides to protect their body and buried their dead.
- *Homo sapiens* arose in Africa and moved across continents and developed into distinct races.
- During ice age between 75,000 – 10,000 years ago modern *Homo sapiens* arose.
- Pre-historic cave art developed about human settlements started.
- The rest of what happened is part of human history of growth and decline of civilisations.

Extra Information on Origin and Evolution of Man



- *Dryopithecus africanus* – is also called as **Proconsul** – discovered from Miocene rocks of Africa and Europe. Fossil ape which believed to be **common ancestor of man and apes** (Gibbons, Orangutan, Chimpanzee and gorilla). It was more ape –like but arms and legs of the same length. Heels in his feet indicate its semierect posture
- *Ramapithecus* fossils were found in **shivalik hills** by **Edward Lewis** (1932) from Pliocene rocks of shivalik hills of India. It was more man like and lived on the tree tops but also walked on the ground. Its jaws and teeth were like those of human modern man.
- *Kenyapithecus wicker* was discovered by **L.S.B, Leaky** (1962) from Pliocene rocks of Kenya (Africa). It was similar to *Ramapithecus* but *Ramapithecus* was older than *kenyapithecus*.
- There is a gap of about 9 to 10 million years in *Ramapithecus* and *Australopithecus* (to be described)
- *Australopithecus africanus* (Africans ape man) – are the ancestors of man **first time stood erect** and showed bipedal movement. Found from Pliocene rocks near Tuang in Africa. Actually skull discovered was 5-6 year old baby so it is also called '**Tuang baby**'.
- In 1981 **Donald Johanson** found a **3.2 million year old skeleton of female human ancestor**. He nicked named it **Lucy** and scientific name is *Australopithecus afarensis*.
- In *Homo habilis* 'habilis' refers to **tool maker**.
- *Homo habilis* discovered by **L.S.B. Leaky** and his wife **Mary Leaky** (1960) discovered the fossils from Pleistocene rocks of Olduvai Gorge in East Africa.
- *Homo erectus* includes three fossils – **Java ape man, Peking man, Heidelberg man**.
- Java man fossils found from Pleistocene rocks in central Java (island of Indonesia)
- Earlier Scientific name of Java man was – *Pithecanthropus erectus* by **Eugene Dubois** 1891.
- Now the scientific name of Java man is - *Homo erectus erectus* by **Mayer** (1980).
- *Pithecanthropus* means **ape man**
- Peking man discovered by **W.C. Pei** (1924) from the lime stone caves of Choukoutien near Peking (Beijing – capital of China was formerly known as Peking) and named as *Sinanthropus*. **David Black** (1927) named it *Sinanthropus pekinesis*. **Mayer** (1950) renamed it *Homo erectus pekinensis* (a subspecies)
- There is another gap of about 50,000 years in the fossil record of evolution of man
- *Homo sapiens neanderthalensis* – Neanderthal valley in Germany from the late Pleistocene
- **Cro-Magnon Man** (*Homo sapiens fossilis*) discovered from Cro-Magnon rocks of France by **MacGregor** in 1868. Emerged about 34000 years ago in Holocene epoch. Thus it is regarded as **most recent and direct ancestor of today's modern man**.
- Cave painting done by Cro – Magnon man
- The blood composition and anatomy of chimpanzee is nearly same as that of man
- Cranial capacity of Gorilla is 500cc, Orang-utan is 375cc, Chimpanzee is 380 and Rhesus monkey – 200 cc.



Schematic representation of Evolution of Man. Age, height and cranial capacity are also given

SUMMARY

The origin of life on earth can be understood only against the background of origin of universe especially earth. Most scientists believe chemical evolution, i.e., formation of biomolecules preceded the appearance of the first cellular forms of life. The subsequent events as to what happened to the first form of life is a conjectured story based on Darwinian ideas of organic evolution by natural selection. Diversity of life forms on earth has been changing over millions of years. It is generally believed that variations in a population result in variable fitness. Other phenomena like habitat fragmentation and genetic drift may accentuate these variations leading to appearance of new species and hence evolution. Homology is accounted for by the idea of branching descent. Study of comparative anatomy, fossils and comparative biochemistry provides evidence for evolution. Among the stories of evolution of individual species, the story of evolution of modern man is most interesting and appears to parallel evolution of human brain and language.